pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Historical Simulation and VaR

WEEK 1 · DAY 2

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Today: the book's VaR, had the past repeated

The book

200 invented bonds, US$490,819,215: synthetic

The curve

U.S. Treasury par yields: real, public domain

The spreads

Course 3's IG and HY indexes: synthetic

Today's question

Had each past day repeated, with today's book held fixed, how bad were the worst days?

And can the answer be checked? A backtest counts the days the VaR was beaten.

Development years only: 2,000 days, 2015 to 2022. The held-out years open in the capstone.

Names for a loss

Every loss today is conditional: had the daily moves of 2015 to 2022 repeated, with the book of September 30, 2026 held fixed.

Allowed names

VaR and expected shortfall

had these days repeated

under these invented inputs

Never called

a forecast or an outlook

an expected move

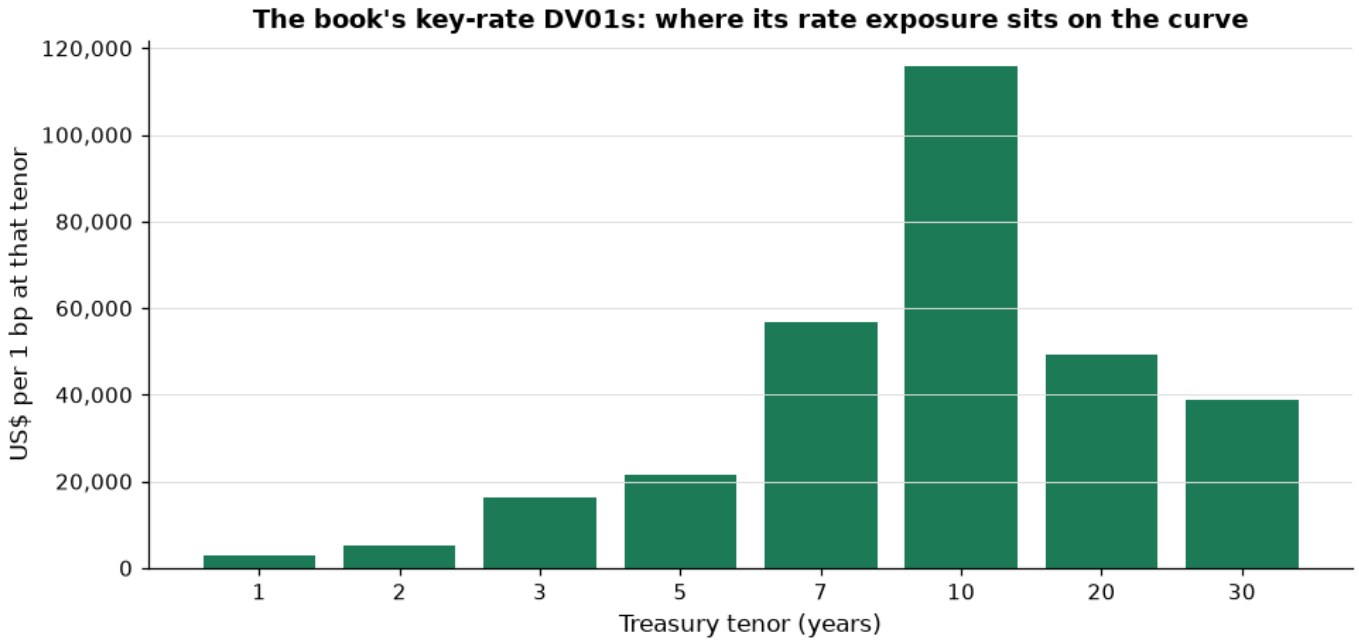
a view on rates or spreads

A VaR is a percentile of a chosen past. Today shows how much that choice matters.

How a bond's P&L works

The problem	A day's moves of the curve and of spreads have to become US$ for each bond
Rates	Minus market value x duration x the Treasury move at the bond's maturity, over 10,000
Spreads	Spreads move in proportion: the bond's OAS x the relative change of its IG or HY index
You control	Duration, spread duration, OAS, maturity, and which index the spread follows
What comes out	The US$ the book makes or loses for a move: DTS, duration times spread

Where the book's exposure sits



US$306,996

per 1 bp of Treasury yield

1% relative widening	US$
High yield	298,157
Investment grade	224,130

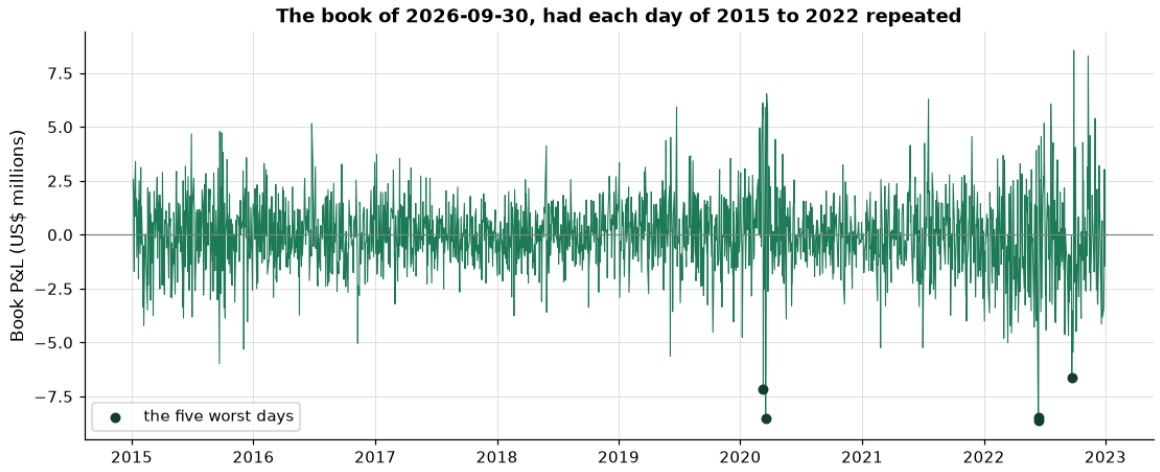
High yield holds less market value; its wider spreads make the same relative move more bp.

Each bond reads the curve at its own maturity, split between the two nearest tenors.

How historical simulation works

The problem	The shape of the next day's moves is unknown; past days are a record of real ones
Scenarios	Each past day's moves are one scenario, applied to today's book
Kept as they were	Fat tails, skew, and how rates and spreads moved together all stay in
You control	Which days, and how a move is applied (first order: duration, DTS)
What comes out	One P&L per past day (historical_pnl); nothing outside those days can appear

The book, had each day repeated



Day	Rates	Spreads	Total (US$)
2022-06-14	-1,681,916.1	-6,952,148.5	-8,634,064.7
2020-03-17	-8,182,034.3	-323,922.5	-8,505,956.8
2022-06-13	-8,393,918.4	-87,817.8	-8,481,736.2
2020-03-10	-6,830,216.5	-328,680.6	-7,158,897.1
2022-09-22	-5,345,683.2	-1,285,052.2	-6,630,735.4

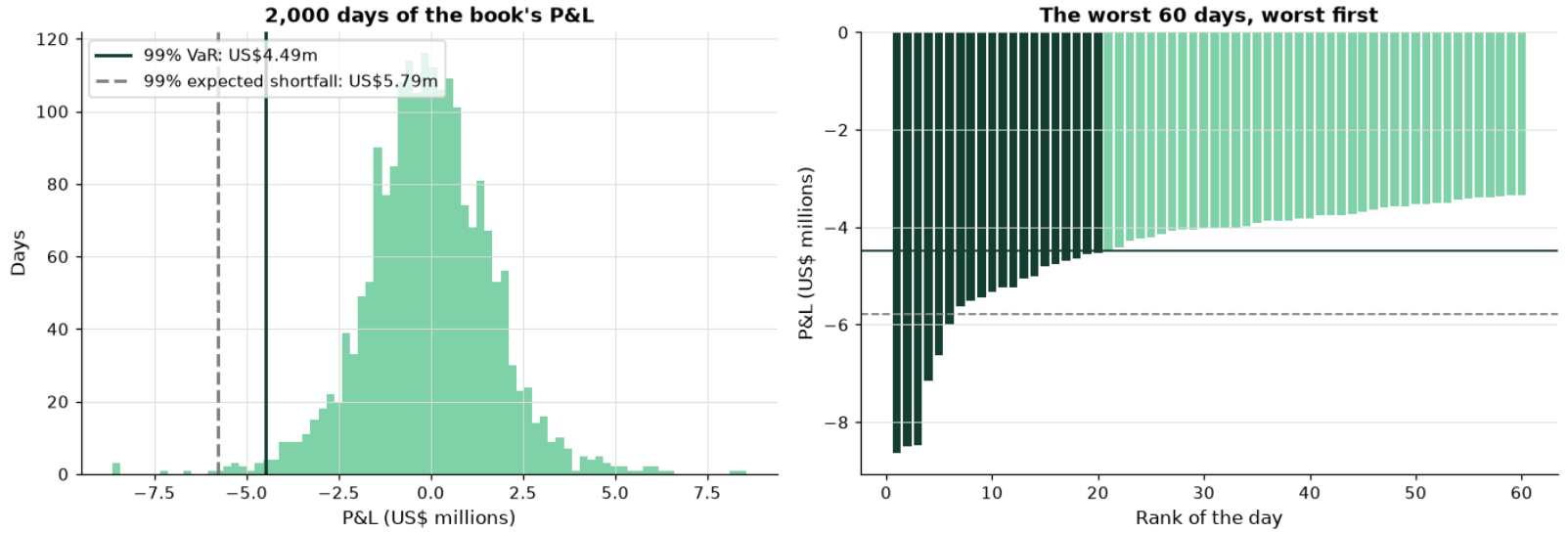
Rates and spreads were nearly uncorrelated (0.01), so the two parts partly offset. Large days came in clusters.

The table says what moved. Why the curve moved is not in these files.

How VaR and expected shortfall work

The problem	A distribution of 2,000 days has to become a number a desk can hold
VaR	The loss exceeded on 1 percent of the days: minus the 1st percentile of the P&L
Shortfall	The average loss on the days at or beyond the VaR: how bad the bad days were
You control	The level (95, 97.5, 99 percent) and how a percentile between two days is read
What comes out	Two US$ losses, the second never below the first (var_es)

VaR and expected shortfall of the book



US$4.49m

99% VaR

US$5.79m

99% expected shortfall

US$2.82m

95% VaR

20

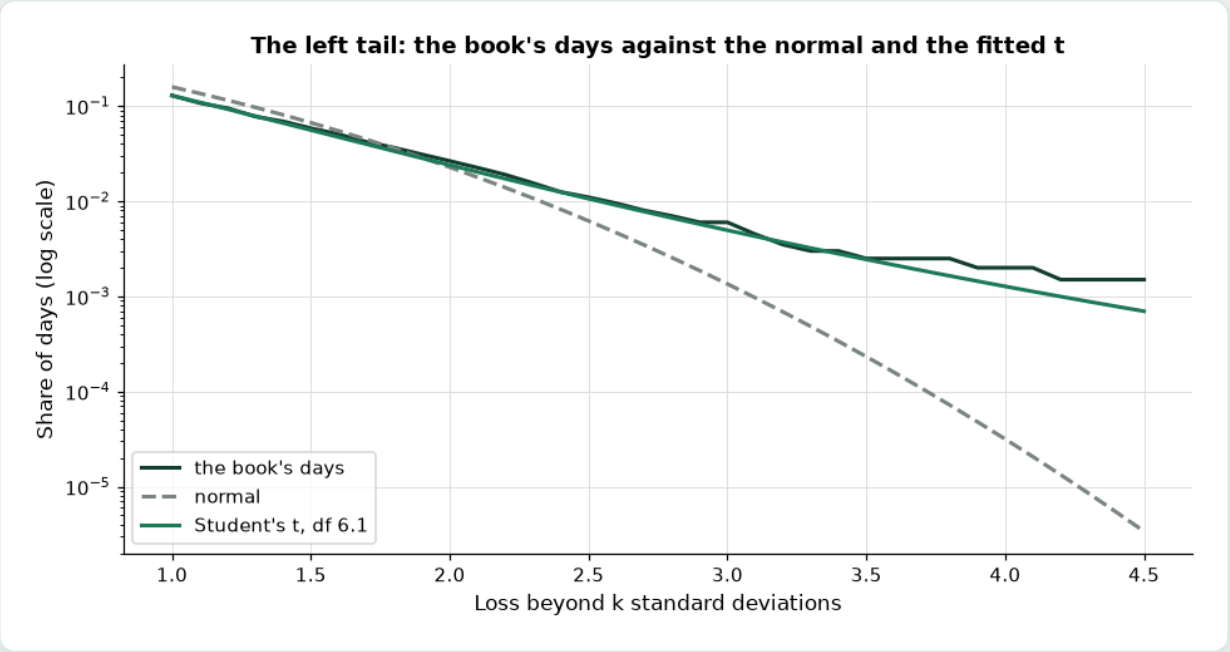
of 2,000 days beyond it

Had 2015 to 2022 repeated, with the book of September 30, 2026 held fixed. A VaR alone hides the worst days.

Three models of the same days

Model	VaR 95%	VaR 99%	ES 99%
historical	US$2.82m	US$4.49m	US$5.79m
normal	US$2.88m	US$4.06m	US$4.65m
t, df 6.1	US$2.76m	US$4.45m	US$5.65m

The normal is close at 95 percent and too small at 99: the book's worst days are more extreme than a normal allows. The fitted t follows the history.



The further into the tail the question, the more the normal model understates.

Every setting of a VaR

Setting	Today's choice	What it controls
level	0.95, 0.975, 0.99	how far into the tail: 1 day in 20, 40, or 100
method	linear (PERCENTILE.INC)	how a percentile between two days is read
window	all 2,000 days; 250, 500, 1,000 in the backtest	how many past days make one VaR
horizon	1 day	the holding period the loss is measured over
model	historical, normal, Student's t	what shape the moves are given

US$4.49m

linear: PERCENTILE.INC

US$4.52m

weibull: PERCENTILE.EXC

US$33,604

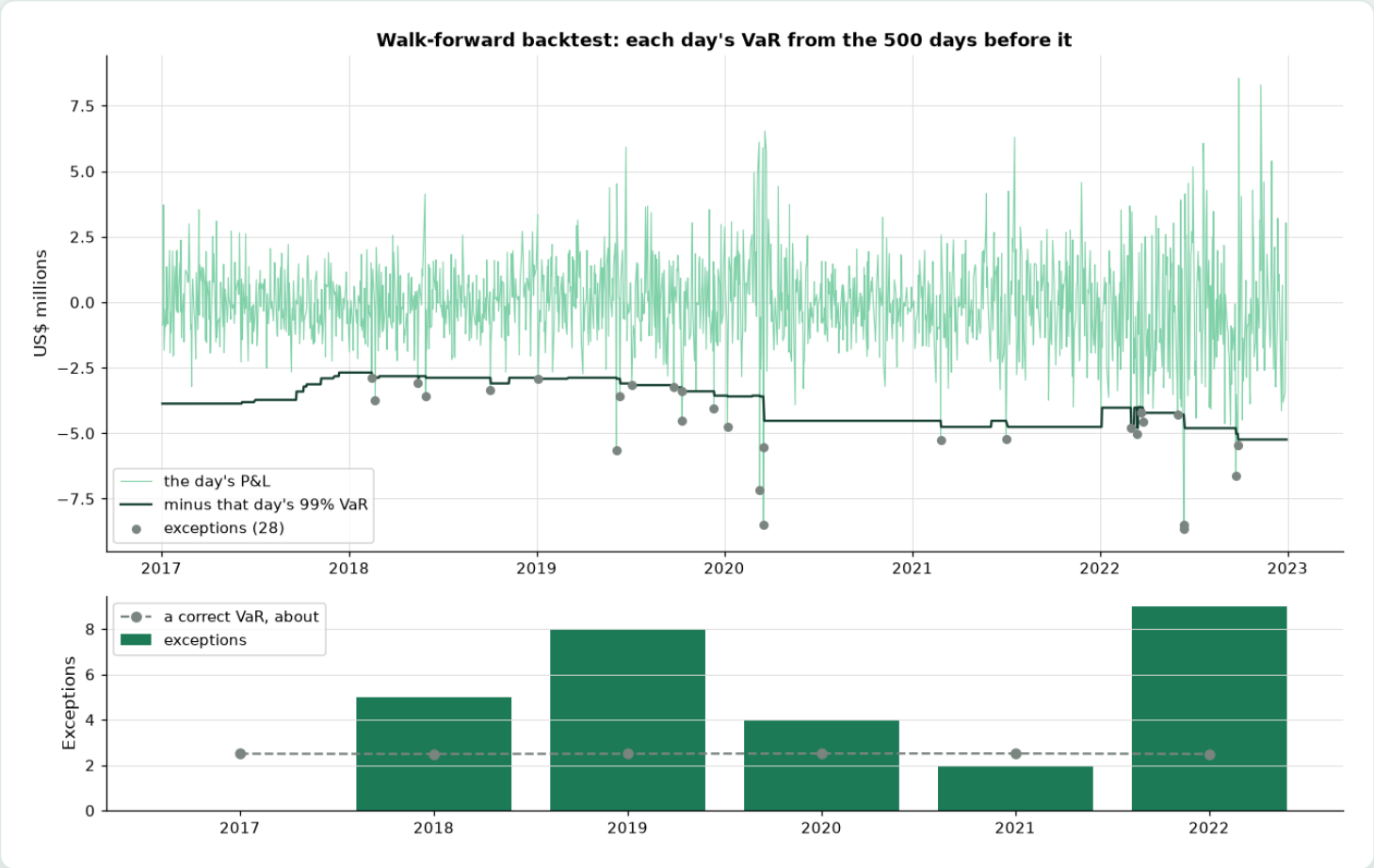
all six methods within

With 2,000 days the methods differ only between the 20th and 21st worst days.

How a backtest works

The problem	A VaR is a claim: a 99 percent VaR should be beaten about 1 day in 100
Walk forward	Each day's VaR from the 500 days before it, never the day itself, then compare
Exceptions	A day whose loss is larger than that day's VaR; correct, they are 1 percent coin flips
Kupiec's test	A likelihood ratio on the count, near a chi-squared with 1 degree of freedom
What comes out	The count, the count a correct VaR gives, and a p-value (kupiec_pof)

The backtest: beaten too often



28

exceptions in 1,500 days

15

for a correct 99% VaR

9.07

Kupiec's statistic

0.0026

p-value

Tested from January 3, 2017. A window of past days is slow: after a calm stretch the VaR is too small for months.

Longer horizons

10-day VaR at 99%	US$	Samples
1-day VaR x square root of 10	14,199,636	2,000
overlapping 10-day sums	15,079,802	1,991
separate 10-day blocks	13,896,841	200

Overlapping sums come out above the square-root rule: large days came in runs, so ten days added up to more than ten independent days would. Separate blocks are independent but few, and their percentile sits between the 2nd and 3rd worst block.

The square-root rule assumes days are independent and alike. These days were not.

RiskMetrics makes VaR a public method

J.P. Morgan and Reuters, Fourth Edition, New York, December 17, 1996

RiskMetrics Technical Document

Launched October 1994; methodology and daily data sets free of charge

Historical simulation "makes no explicit assumptions about the distribution of asset returns". And on the first page: "no amount of sophisticated analytics will replace experience and professional judgment in managing risks."

Source: RiskMetrics Technical Document, Fourth Edition (1996): cover, preface page iii, pages 6 and 27; msci.com. Backtest: P. H. Kupiec, FEDS 95-24, Federal Reserve Board, 1995.

Excel side by side

Excel side by side (tested in Excel)

`=-PERCENTILE.INC(B2:B2001,0.01)`	99% VaR US$4,490,319
`=-AVERAGEIF(B2:B2001,"<="&PERCENTILE.INC(B2:B2001,0.01))`	99% shortfall US$5,790,757
`=-(AVERAGE(B2:B2001)+NORM.S.INV(0.01)*STDEV.S(B2:B2001))`	normal model US$4,061,269
`=-PERCENTILE.EXC(B2:B2001,0.01)`	PERCENTILE.EXC US$4,523,251
`=SUMPRODUCT(--(-F2:F1501>G2:G1501))`	28 exceptions
`=CHISQ.DIST.RT(K3,1)`	Kupiec p-value 0.0026

B2:B2001 holds the 2,000 days of P&L; F and G the tested days and their VaR. Every value equal to Python.

Save the day: search what Git tracks

1. Init

The work folder, local settings, the guard, .gitignore

2. Commit

mlkit.py, riskkit.py, and its tests

3. git grep

Search the tracked files only: every def in riskkit.py

```
git grep -n "^def " -- riskkit.py
```

6 functions: day 1's three and today's three, each with its line number.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a tail taken below the VaR: VaR US$4,490,319, and a "shortfall" that is a gain of US$12,175.

Code that runs is not the same as code that is right.

What you can do now

- Measure exposure by key rate and by DTS
- Apply past days to today's book (historical_pnl)
- Read VaR and expected shortfall (var_es)
- Set the normal and a fitted t beside the history
- Backtest and test the count (kupiec_pof); 13 tests pass

Week 1, Day 3 Correlated defaults and the loss distribution